

AVIRAL GARG

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EDUCATION

M.S., Computer Science — University of Kentucky, Lexington, Kentucky	Jan 2024 – May 2026
B.Tech., Electronics & Computers Engineering — Thapar Institute of Engineering & Technology, India	May 2018 – Jun 2022

CERTIFICATIONS

IBM RAG and Agentic AI Professional Certificate — IBM / Coursera	Aug 2026
Machine Learning A–Z: AI, Python & R — Udemy	Sep 2023
Mastering Data Structures & Algorithms in C/C++ — Udemy	Jul 2023
Python 3.11 — Udemy	May 2023
Java for Beginners — Coursera	Sep 2023

TECHNICAL SKILLS

Languages: Python, C++, C, C#, JavaScript, SQL, HTML5/CSS3, Bash
AI/ML: PyTorch, TensorFlow, Scikit-learn, Transformers, BERT, Deep Reinforcement Learning (DQN, PPO), Hugging Face, NLP
GenAI & Agentic Systems: RAG, LLMs, LangChain, LangGraph, CrewAI, AutoGen, BeeAI, Model Context Protocol (MCP), Vector Databases (FAISS, Chroma), Prompt Engineering
Cloud & Systems: AWS, Docker, Kubernetes, Knative, Serverless Architecture, Distributed Systems, Linux, Network Security
Web, Full-Stack & Data: React, React Native, Flask, .NET, Firebase, PostgreSQL, MySQL, MongoDB, NumPy, Pandas

PROFESSIONAL EXPERIENCE

Student Engineering Technician — University of Kentucky	Feb 2026 – May 2026
- Built Python automation pipelines and a custom GUI, standardizing QA/QC workflows across 45+ sensors and controllers spanning 3 research systems, holding temperature error margins within 0–1°. - Led a 5-person engineering team and compiled system diagrams and validation results supporting a peer-reviewed ASABE technical publication.	
Front End Developer — Dept. of Computer Science, University of Kentucky	Sep 2024 – Jan 2025
- Built full-stack UI/UX for GenAI and RAG applications using LlamaIndex and FAISS vector databases, deploying a Gradio dashboard across university servers to a large concurrent user base. - Integrated DBMS architecture and NLP-driven data-cleaning pipelines, eliminating indexing redundancies.	
Software Engineer, R&D — Recorders & Medicare Systems, India	Jul 2022 – Jun 2023
- Designed full-stack .NET interfaces for clinical diagnostic hardware (EEG, ECG, X-ray, BP monitors) deployed at scale in zero-error clinical environments. - Engineered a service-oriented architecture for multi-VM deployment; integrated SQL/MongoDB for embedded medical data storage.	

PROJECTS

Risk-Aware AI Portfolio Management System — Deep RL + LLM Auditor	Jan – May 2026
Built a PPO reinforcement-learning trading agent with an LLM risk auditor that cross-checks allocations against live financial news, fusing quantitative and qualitative signals for interpretable, risk-aware decisions.	
Dynamic Portfolio Management with Bandit Algorithms — ML	Aug – Dec 2025
Benchmarked LinUCB and PPO allocation strategies across SPY, QQQ, TLT, and GLD, evaluating Sharpe ratio and max drawdown under realistic transaction costs.	
Knative Function-Chain Autoscaling & Stateful File Server — Cloud/Systems	Aug – Dec 2025
Compared concurrency-, RPS-, and position-aware autoscaling policies for chained functions on Kubernetes/Knative; built a Sun RPC file server in C with per-user quotas.	
Fake News Detection on Social Media — NLP	Jan – May 2025
Fine-tuned BERT against SVM/Naive Bayes baselines to classify misinformation across 40,000+ labeled samples using custom scraping and NLP preprocessing pipelines.	